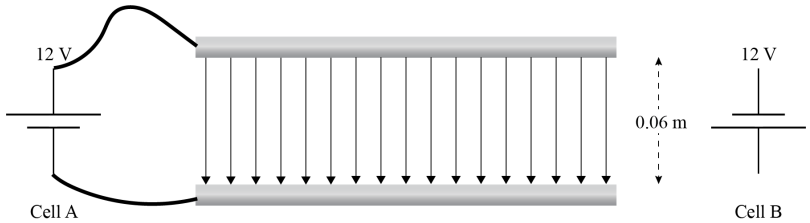
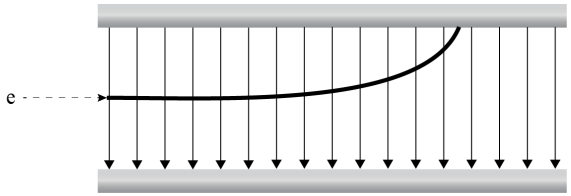
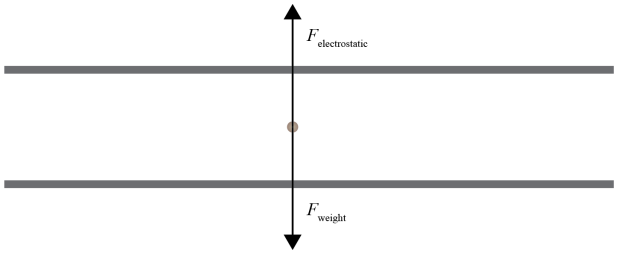
Assessment Schedule – 2024

Physics: Demonstrate understanding of electricity and electromagnetism (91173)

Evidence Statement

Q	Evidence	Achievement	Merit	Excellence
ONE (a)		• Positive terminal connected to the top plate.		
(b)	$E = \frac{V}{d} = \frac{12}{0.06} = 200 \text{ V m}^{-1}$ $E_p = Eqd = 200 \times 1.6 \times 10^{-19} \times 0.06 = 1.92 \times 10^{-18} \text{ J}$ $E_p = \frac{1}{2}mv^2 \Rightarrow 1.92 \times 10^{-18} = \frac{1}{2} \times 9.1 \times 10^{-31} v^2$ $\Rightarrow v = \sqrt{4.22 \times 10^{12}} = 2.1 \times 10^6 \text{ m s}^{-1}$	• E = 200 V/m OR Uses $E_p = E_k$	• $E_p = 1.92 \times 10^{-18}$ J OR Finds v with error.	• 2.1×10^6 m s^{-1}
(c)(i)		• Any one of: - curve upwards - constant force - uniform field - parabolic path.	Curve upwards And one of - parabolic path - accelerates up - constant force up.	
(ii)	The electron experiences an electric force away from the negative plate. The electric force is constant as the electric field is constant. This makes the electron follow a parabolic path.			

(d)(i) (ii)	 $mg = Eq = \frac{V}{q}d \Rightarrow 1.95\times10^{-15}\times9.8 = \frac{6.54\times10^{2}}{0.0219}q$ $q = 6.4\times10^{-19}$ $ne = q \Rightarrow n = \frac{q}{e} = \frac{6.4\times10^{-19}}{1.6\times10^{-19}} = 4$ Note: $E = \frac{V}{d} = \frac{6.54\times10^{2}}{0.0219} = 29\,900$ V m^{-1}	• Two force arrows. OR $E = 29\,900$ V m^{-1} OR Finds $mg = 1.95\times10^{-15}\times9.8$ $= 1.91\times10^{-14}$ N	• Found q. OR Arrows and found E. OR Arrows and divided any q by electron.	• Correct answers.

NØ	N1	N2	A3	A4	M5	M6	E7	E8
No evidence	NA	1a or1m	2a or 1 a +1m	3a	2m+1a	2m+2a or 3m	1e+1m+2a or 1e +2m	2e

Q	Evidence	Achievement	Merit	Excellence
TWO (a)	The voltage is the same across both branches. The branch with B and C has two lamps in series so double the resistance – half the current. Did not accept current is 1.4 A and each lamp gets half of it.	• Double the resistance. OR Voltage is the same across both branches / parallel.	Full argument.	
(b)	Circuit Current = 1.4 + 0.7 = 2.1 A Voltage across the resistor = $IR = 2.1 \times 3.5 = 7.35$ V Voltage across parallel branch = 12 – 7.35 = 4.65 V Resistance lamp D = $R = \frac{4.65}{1.4} = 3.3\,\Omega$ OR Circuit resistance: $R = \frac{V}{I} = \frac{12}{2.1} = 5.71 \Rightarrow$ R of parallel = 5.71 – 3.5 = 2.21 Ω And $\left(\frac{1}{2x} + \frac{1}{x}\right)^{-1} = 2.21$ so x = 3.3 Ω or equivalent	• Total current = 2.1 A	• 2.21 Ω OR 4.65 V	• Show 3.3 Ω

(c)	• Total R increases. • Less current flows in circuit and through the resistor. • Total V constant. • Voltage across the resistor drops. • Power determines brightness. • Lamp D gets more V and is brighter. OR Power at start in lamp D is: $V = IR = 1.4 \times 3.3 = 4.6$ V, $P = IV = 4.6 \times 1.4 = 6.4$ W Power of D after lamp B stops: Total $R = 3.3 + 3.5 = 6.8\,\Omega$ $I = \frac{12}{6.8} = 1.8$ A $V = IR = 1.8 \times 3.3 = 5.9$ V $\Rightarrow P = IV = 5.9 \times 1.8 = 10.6$ $= 11$ W Hence, lamp D is now brighter.	• Total R increases. OR Recognition circuit is now series. OR 4.6W OR $R = 6.8\,\Omega$	• Two correct linked statements. OR P = 6.4 W and 5.9 V. Incorrect statement about R, but linked with argument why Lamp D is dimmer.	• 6.4 W and 11 W. OR • Full answer.
(d)	In Lamp D: $530 / \text{minute} \Rightarrow \frac{530}{60} = 8.83\ \text{J s}^{-1} = = 8.8\ \text{W} = P$ $P = I^2R \Rightarrow I = \sqrt{\frac{P}{R}} = \sqrt{\frac{8.83}{3.3}} = 1.6$ A Total current $= 1.5 \times 1.6 = 2.5$ A	• Finds 8.8 W.	• Correct power and 1.5 times any calculated current. OR Finds 1.6 A.	• 2.5 A

NØ	N1	N2	A3	A4	M5	M6	E7	E8
No evidence	NA	1a or1m	2a or 1 a +1m	3a	2m+1a	2m+2a or 3m	1e+1m+2a or 1e +2m	2e

Q	Evidence	Achievement	Merit	Excellence
THREE (a)	Right end clearly labelled negative.	• Right labelled negative.		
(b)	$V = BvL = 31\times10^{-6}\times4.1\times9.8 = 1.2(4)\times10^{-3}$ V	• $1.2(4) \times 10^{-3}$ V		
(c)	The metal sprayer boom contains electrons which have a negative charge. The electrons move with the boom cutting the magnetic field. The electrons experience a force which causes charge separation, and this is the induced voltage.	• ONE correct point.	• TWO correct linked points.	• Full answer.
(d)(i) (ii)	The direction of the force is clearly indicated with a downwards arrow. $P = IV \Rightarrow 3 = 6I \Rightarrow I = 0.5$ A For one lamp: Force on the wire $= BIL = 31\times10^{-6}\times0.5\times9.8 = 1.5\times10^{-4}$ N OR $P = \frac{V^2}{R} \Rightarrow R = \frac{V^2}{P} \Rightarrow R = \frac{6^2}{3} = 12\ \Omega$ Current in the circuit $= \frac{12}{24} = 0.5$ A	• Finds 0.5 A. OR Arrow downwards showing force direction. OR Finds $R = 12\ \Omega$.	• Correct direction **and** finds 0.5A or 12 Ω. OR Finds 1.5×10^{-4} N.	• Correct answer and direction.

NØ	N1	N2	A3	A4	M5	M6	E7	E8
No evidence	NA	1a or1m	2a or 1 a +1m	3a	2m+1a	2m+2a or 3m	1e+1m+2a or 1e +2m	2e

Cut Scores

Not Achieved	Achievement	Achievement with Merit	Achievement with Excellence
0 – 07	08 – 13	14 – 18	19 – 24